Segfault when running certain circuits with the GPU statevector simulator,

https://github.com/Qiskit/qiskit-aer/issues/1647

ROW 11

Segfault when running certain circuits with the GPU statevector simulator, `AerPauliExpectation`, and certain noise models #1647

New issue

Closed

HaoTy opened on Nov 6, 2022 · edited by HaoTy

Edits

Informations

- **Qiskit Aer version:**

```
qiskit                  0.39.2
qiskit-aer              0.11.1
qiskit-aer-gpu          0.11.1
qiskit-ibmq-provider    0.19.2
qiskit-optimization     0.4.0
qiskit-terra            0.22.2
```

- **Python version:** 3.9.12
- **Operating system:** CentOS Stream 8
- **GPU:** A100-SXM4-80GB
- **CUDA version:** 11.7.0

What is the current behavior?

Running certain circuits (e.g. the QAOA expectation value of an 18-qubit `SKModel` instance, see below) with the GPU statevector simulator, `AerPauliExpectation`, and certain noise models (e.g. 2-qubit depolar error) causes segmentation fault.
Using the CPU statevector simulator does not have this issue.
Some other circuits, for example, the QAOA expectation value of the `Maxcut` problem, do not have this issue.
Some other noise models, including ideal simulations, Pauli errors, and 1-qubit depolar errors, do not have this issue.
16 qubits or below do not have this issue (the exact number may depend on machine specs).

Steps to reproduce the problem

```
import numpy as np
import networkx as nx
from qiskit_aer import AerSimulator
from qiskit_aer.noise import NoiseModel
from qiskit_aer.noise.errors.standard_errors import depolarizing_error
from qiskit.algorithms import QAOA
from qiskit.algorithms.optimizers import SPSA
from qiskit.utils import QuantumInstance
from qiskit.opflow import AerPauliExpectation
from qiskit_optimization.applications import Maxcut, SKModel
from qiskit.utils import algorithm_globals
from docplex.mp.model import Model

n = 18
p = 1
seed = 0

# Maxcut does not produce segfault
# graph = nx.random_regular_graph(3, n, seed)
# problem = Maxcut(graph).to_quadratic_program()

# SKModel with n=18 or more produces segfault
problem = SKModel(n, seed).to_quadratic_program()

# Some other problems can also cause this issue
# number_set = np.random.default_rng(seed).integers(2 ** (n // 2), size=n) + 1
# mdl = Model(name="Number partitioning")
# x = {i: mdl.binary_var(name=f"x_{i}") for i in range(n)}
# mdl.minimize(mdl.sum(num * (-2 * x[i] + 1) for i, num in enumerate(number_set)) ** 2)
# problem = from_docplex_mp(mdl)

H, offset = problem.to_ising()
```

Assignees

doichanj

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants

```
# x = {i: mdl.binary_var(name=f"x_{i}") for i in range(n)}
# mdl.minimize(mdl.sum(num * (-2 * x[i] + 1) for i, num in enumerate(number_set)) ** 2)
# problem = from_docplex_mp(mdl)

H, offset = problem.to_ising()

noise_model = NoiseModel()
noise_model.add_all_qubit_quantum_error(depolarizing_error(0.01, 2), 'cx')
algorithm_globals.random_seed = seed
backend = AerSimulator(method="statevector", device="GPU", noise_model=noise_model)
quantum_instance = QuantumInstance(
    backend=backend,
    seed_simulator=seed,
    seed_transpiler=seed,
)

algorithm = QAOA(
    SPSA(),
    reps=p,
    quantum_instance=quantum_instance,
    expectation=AerPauliExpectation(),
)
algorithm._check_operator_ansatz(H)
energy_evaluation= algorithm.get_energy_evaluation(H)
parameters = 2 * np.pi * np.random.default_rng(seed).random(2 * p)
energy_evaluation(parameters)
```

What is the expected behavior?

The simulator should be able to evaluate the QAOA energy without causing segfault.

HaoTy added bug on Nov 6, 2022

hhorii assigned doichanj on Nov 7, 2022

doichanj on Nov 7, 2022 Collaborator

This issue is caused by gate fusion. Some operations inserted by noise model are fused and fused unitary matrix is larger than matrix buffer allocated on GPU. I will fix this issue by searching max matrix qubits after fusion
Until the fix will be released, please disable fusion by setting `fusion_threshold=30` (larger than circuits qubits)

doichanj mentioned this on Nov 7, 2022

Fix for GPU simulator, memory allocation #1650

HaoTy on Nov 8, 2022 Author

Thanks for the quick help! I can confirm that disabling fusion is an effective workaround.

hhorii on Nov 15, 2022 Collaborator

I would like to create a new issue for segmentation faults due to massive cp gates.

hhorii closed this as completed on Nov 15, 2022

hhorii mentioned this on Nov 15, 2022

Segmentation faults from massive CP gates #1657